

Q When can we see the much anticipated JavaScript 2.0? Could we say that for Web developers, this new version will be worth waiting for?

We had already kind of created what JavaScript 2 eventually became, which was ES4. It just had to be abandoned as that fourth edition, and its best parts carried forward into ES6, now called ES2015. The ECMA-262 standard has now moved to an yearly release cadence, so there's no need to await a 'big bang' 2.0 that might not ever have come out under such an 'all-or-nothing' approach. This more closely matches how software is developed and tested at scale.

Q A year back, there was an announcement on WebAssembly, which you had also called a 'game-changer'. Can you tell us about the status of the project and the support from browser vendors?

I helped announce WebAssembly, which is directly descended from asm.js, the statically typed subset of JavaScript that can be optimised to near-native code speed on par with Google's Portable Native Client (PNaCl) and other such predecessors. Since then, all four top browser vendors have moved forward with WebAssembly, and just the other week, three of them announced their plans to release WebAssembly support, which is likely to go live in the first quarter of 2017.

Q The ECMA TC39 committee that develops and standardises the ECMAScript (JavaScript) has recently become open to community contributions. Can organisations from India join and contribute towards language specifications by leveraging this latest development?

ECMA still requires people to agree to give up ownership of intellectual property according to its 'Code of Conduct in Patent Matters', which has language for ECMA members and non-members alike. Still, having worked in TC39 for over a decade since helping reform it once Firefox had restarted the browser market, I think it always helps to have a corporate or not-for-profit organisation that can join ECMA.

Q Years after introducing the world to JavaScript, you co-founded the Mozilla project. Why was there a need for launching Mozilla?

I had finished JavaScript 1.0 and standardised it with excellent colleagues from IBM, Microsoft, Sun and other companies, as ECMA-262 Edition 1. Netscape had finally invested in a JavaScript team, led by Clayton Lewis, with outstanding programmers who could carry on the work both in the engine I had rewritten (SpiderMonkey; the first engine was called Mocha), and in ECMA TC39. I was ready for something new.

Because Netscape was being driven out of business due to Microsoft's abuse of its Windows OS monopoly on personal computers, Netscape's executives and top engineers

“ Even before Mozilla or an open source implementation of JavaScript, I used all the early adopter support techniques and energy that I had acquired over the years in software, going back to Silicon Graphics. ”

were warm to the idea of releasing the Netscape code as open source. This was a daunting task (free the lizard!) and the first big commercial, closed-source conversion to open source.

Moreover, the chance to be the chief architect of the mozilla.org project—to work from the start on an 'escape pod' for the browser, was one I could not pass up. It is hard to remember how unlikely the later success of Firefox seemed. But even in 1998, some of us believed that Mozilla could bring back the browser market from its then-impending inevitable monopolisation by Internet Explorer.

Q The original Mozilla project brought in Mozilla Corporation as a for-profit arm to support the Mozilla Foundation. Is it easy to garner profits with open source developments?

Initially, mozilla.org had no legal structure. It was a 'virtual organisation' inside Netscape, soon bought by AOL. But in 2003, we spun Mozilla out from AOL as a non-profit, the Mozilla Foundation, with no for-profit subsidiary. After some promises from the US IRS were broken, we ended up creating the Mozilla Corporation as a wholly-owned subsidiary of the Mozilla Foundation.

I cannot say this model is easy to transfer to other open source projects. It is hard, perhaps even impossible, as Mozilla Firefox was *sui generis* (unique). What I have seen work for other open source projects ranges from commercial dual-licence to indie-developed or consulting-funded and to SaaS subscriptions. No one has cracked the code on how to make predictable money from open source. We do know that creating great software can create even greater value through ecosystem effects such as easier integration, community- and partner-built extensions and standards for interoperation.

Q How is the Indian developer base contributing to Mozilla's open source solutions?

I am no longer at Mozilla and cannot say much on that topic, since I am busy with Brave, which is doing well around the world, including in India. We already have several volunteer Brave Squads around the country. Some of them are from Lucknow, Hyderabad and even Jaipur. They are active in fixing bugs for Brave as well as in localisation efforts (adding Hindi and Bengali). Developers can get involved via the official Brave repository on GitHub or by joining us @BeBraveIndia on Twitter, as well as following the Facebook group.